

DevOps Lab

Program 2- Creating a Multi-Stage Dockerfile

Project Structure:

```
program_2
  Dockerfile
  package.json
  package-lock.json
  node_modules/
  src/
  index.js
```

Step 1: Create Project Folder

```
mkdir program_2
cd program_2
```

Step 2: Initialize Node.js Project

```
npm init -y
```

Step 3: Install Express Framework `npm install express`

Step 4: Write the DockerFile code and the [index.js](#) code

Step 5: Build the Docker Image `docker build -t program2-app .`

```
srikanths@Srikanths-MacBook-Air ~/D/H/R/f/program4 (main)> cat index.js
const express = require('express')
const app = express();

app.get('/', (req, res)=>{
  res.send("Hello World")
})
const PORT=8000;
app.listen(PORT, ()=>{
  console.log("Listening on port 8000")
})
srikanths@Srikanths-MacBook-Air ~/D/H/R/f/program4 (main)> docker build -t program4 .
[+] Building 5.7s (11/11) FINISHED
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 225B
=> [internal] load metadata for docker.io/library/node:20-alpine
=> [internal] load .dockerignore
=> => transferring context: 28
=> [1/6] FROM docker.io/library/node:20-alpine@sha256:6178e78b972f79c335df281f4b7674a2d85071aae2af020ffa39f0a770265435
=> => resolve docker.io/library/node:20-alpine@sha256:6178e78b972f79c335df281f4b7674a2d85071aae2af020ffa39f0a770265435
=> [internal] load build context
=> => transferring context: 3.85MB
=> CACHED [2/6] WORKDIR /app
=> [3/6] COPY package*.json .
=> [4/6] RUN npm install express
=> [5/6] COPY . .
=> [6/6] RUN npm install
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:dd3d1f2fbb7e85ce4d1e4db91264b3561d50756b8da79de5a4393de46bc49ff
=> => exporting config sha256:1ab999ddcfd04c518a3d1d5039a339116dd825853f64c5bd42f2ed1236bb6f1
=> => exporting attestation manifest sha256:c5afac97944b0c0655d12ff33c3959c32b171d8555b7a8195b4c9affec22300c
=> => exporting manifest list sha256:5e9ab0e9d20fd9198deabd34e5ebe4f00e747c5721376be82af2d0338ed902aa
=> => naming to docker.io/library/program4:latest
=> => unpacking to docker.io/library/program4:latest
View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/tvx5g6hj0yrmc9p5104bc6p49
srikanths@Srikanths-MacBook-Air ~/D/H/R/f/program4 (main)> docker run -d -p 8000:8000 program4
b01c7364d00702e100c020a8b8c70b031f7bdcde82952eda4fcfb861db5355e0
srikanths@Srikanths-MacBook-Air ~/D/H/R/f/program4 (main)>
```

Step 6: Verify the output on localhost:8000

